INCH-POUND

MIL-E-85082/5

16 November 1992

MILITARY SPECIFICATION SHEET

ENCODER, SHAFT POSITION TO DIGITAL
CONTACT TYPE, ABSOLUTE
TYPE 11KPL13R08A

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Encoders, Shaft Angle to Digital, described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) cited in the solicitation: MIL-E-85082.

TABLE I. Requirements.

Requirements	Value	Unit	Tolerance
Code and scan	Gray, Lead-Lag		
Capacity	2^{13}	counts	maximum
Resolution	2^{8}	counts/turn	
Full count revolutions	32	turns	
Voltage interrogation	12	volts DC	maximum
Contact current	2.0	mA/bit	maximum
Starting torque	0.5	oz-in	maximum
Running torque	---	oz-in	maximum
Operating speed	200	rpm	maximum
Slew speed	1000	rpm	maximum
Radial play	0.0010	inches	maximum
End play	0.0020	inches	maximum
Total shaft runout	0.0010	inches	maximum
Friction torque	0.5	oz-in	maximum
Transition accuracy	±1/2	LSB	
Operating temperature	-25° to +70°	degrees C	min-max
Moment of inertia	0.17	$oz\text{-}in^2$	maximum
Weight	6.0	oz	maximum
Life	5,000,000	revolutions	minimum

AMSC - N/A FSC 5990

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TABLE II. Part identifying number variant characteristics.

Part Identifying No. 1/	A ±0.020	E +0.000 -0.001	G Maximum	Shaft Style	Terminal End	U +0.0000 -0.0001
M85082/5-01A	.620	.440	2.500	Slotted	Wire Leads	.1248

1/ Part identifying number M85082/5-01A shall be the NATO standard.

1/ The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation.

TABLE III. Wire termination identification.

FUNCTION	LAG	LEAD
2^0	WHITE/BLUE/BLACK	
2^1	BROWN	WHITE/BROWN
2^2	RED	WHITE/RED
2^3	ORANGE	WHITE/ORANGE
2^4	YELLOW	WHITE/YELLOW
2^5	GREEN	WHITE/GREEN
2^6	BLUE	WHITE/BLUE
2^7	VIOLET	WHITE/VIOLET
2^8	GRAY	WHITE/GRAY
2^9	WHITE	WHITE/BLACK
2^{10}	ORANGE/BLACK	WHITE/ORANGE/BLACK
2^{11}	YELLOW/BROWN	WHITE/YELLOW/BROWN
2^{12}	BROWN/RED	WHITE/BROWN/RED
	+12VDC	RED
	COMMON	BLACK

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NOTES:

1. Dimension H shall be 0.030 with a tolerance of ±0.001.
2. Dimension I shall be 0.062 with a tolerance of +0.000/-0.005.
3. Dimension J shall be 0.090 with a tolerance of +0.005/-0.000.
4. Rotation for increasing count shall be clockwise.
5. Lead exit shall be sealed.
6. Radial and end play shall be performed with 4 oz. reversing load maximum applied 0.12 inch from the end of the shaft.
7. Wire leads shall be in accordance with MIL-W-16878/4 size 28 and shall be 3.5 ±.06 inches long.
8 The recommended resistive load is 0.5K ohms.

Custodians:
Navy - AS
Army - AR
Air Force - 85

Review activities:
DLA - ES

Preparing activity:
Navy - AS

(Project 5990-0405-5)

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